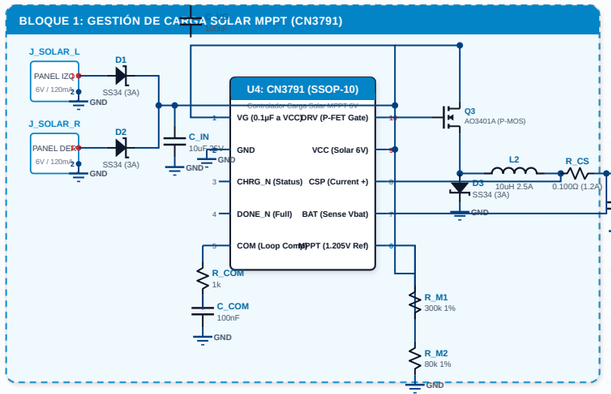
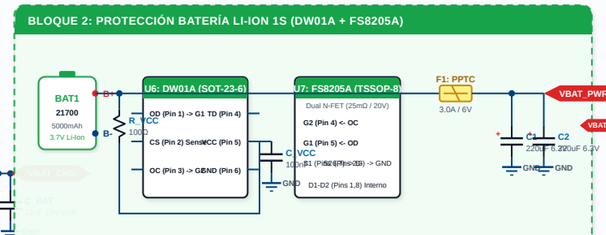
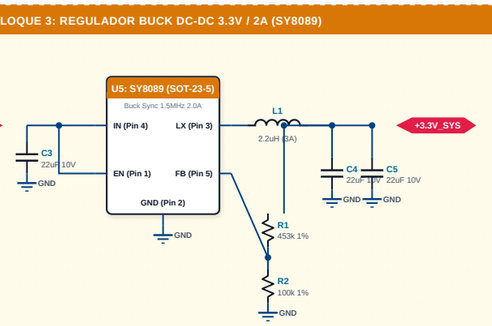
BLOQUE 1: GESTIÓN DE CARGA SOLAR MPPT (CN3791)



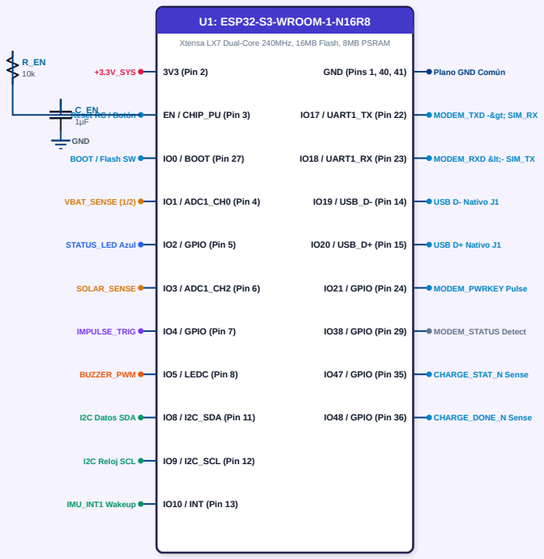
BLOQUE 2: PROTECCIÓN BATERÍA LI-ION 1S (DW01A + FS8205A)



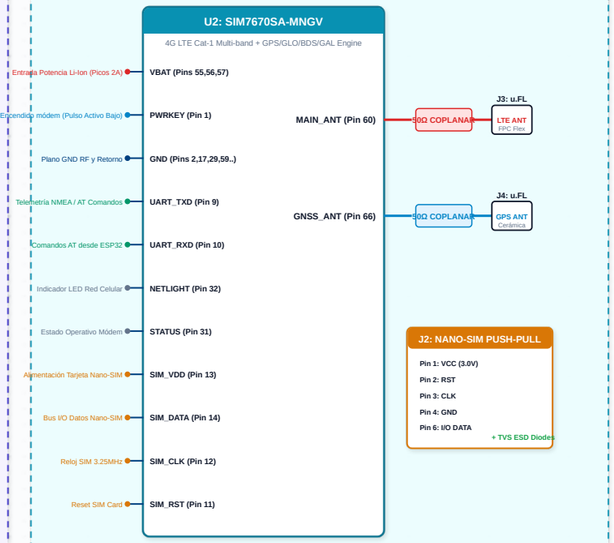
BLOQUE 3: REGULADOR BUCK DC-DC 3.3V / 2A (SY8089)



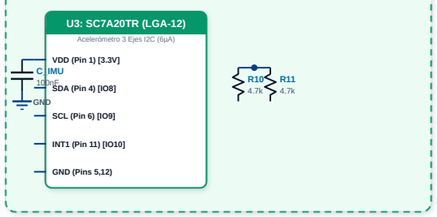
BLOQUE 4: MICROCONTROLADOR PRINCIPAL (ESP32-S3-WROOM-1-N16R8)



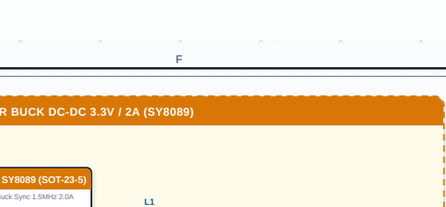
BLOQUE 5: TELEMETRÍA MÓVIL 4G LTE CAT-1 + GNSS (SIM7670SA)



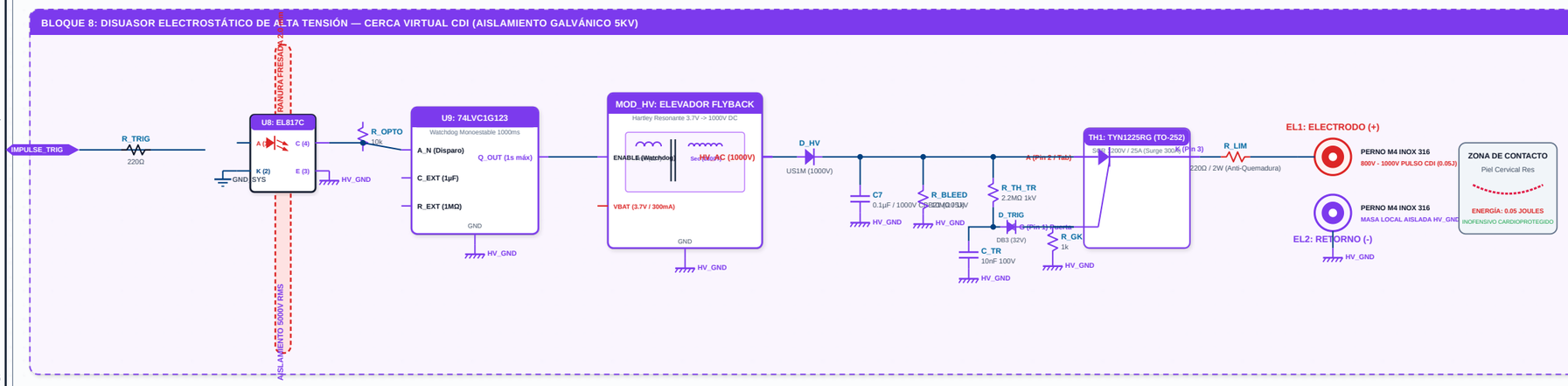
BLOQUE 6: SENSOR INERCIAL / PODÓMETRO GANADERO (SC7A20)



BLOQUE 7: ALARMA ACÚSTICA DISUASORIA (BUZZER 95DB)

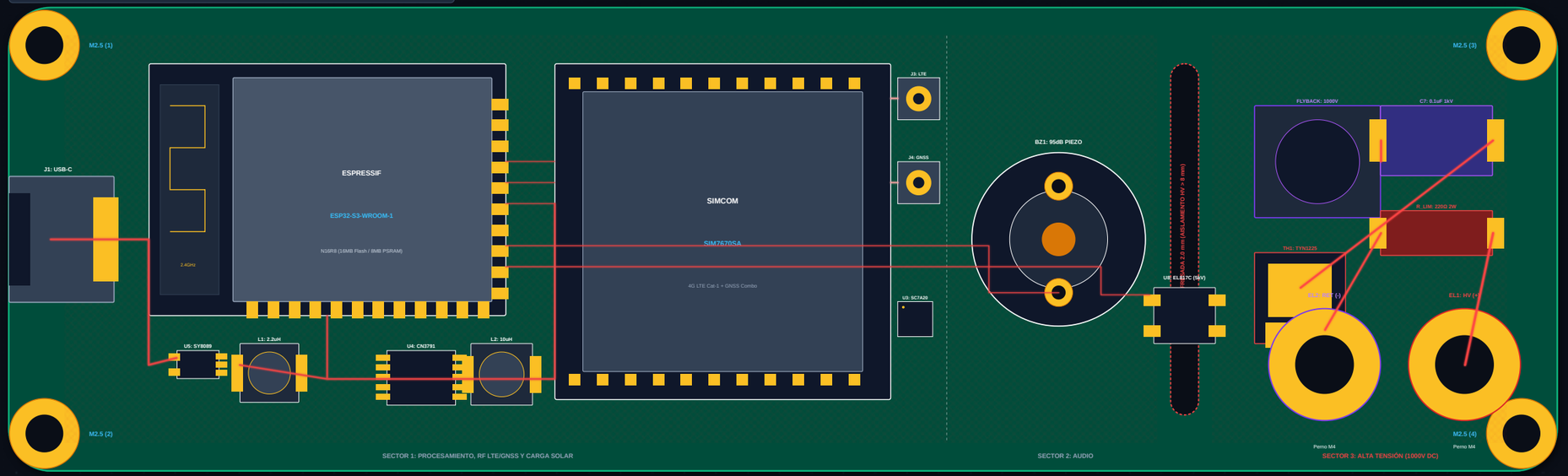


BLOQUE 8: DISUASOR ELECTROSTÁTICO DE ALTA TENSIÓN — CERCA VIRTUAL CDI (AISLAMIENTO GALVÁNICO 5KV)



COW-AI TECHNOLOGIES S.A.	PROYECTO:	REVISIÓN:
DIAGRAMA DEL CIRCUITO COMPLETO ALL-IN-ONE (8 BLOQUES)	COLLARNET V2	2.2
FECHA: 2026-10-27 | DISEÑADOR: ING. HARDWARE CDMX	NORMA: IPC-2221 / IEC 60664-1 (5kV)	PLANO 1 DE 6

VISTA 1: CARA SUPERIOR (TOP LAYER) — PISTAS DE COBRE ROJO, PADS SMT Y SERIGRAFÍA



VISTA 2: CARA INFERIOR (BOTTOM LAYER) — PISTAS AZULES, CLIPS BATERÍA 21700 Y NANO-SIM



PCB STACKUP & SPECIFICATIONS

ESPECIFICACIÓN FÍSICA:

• Dimensiones:	**110.60 x 33.10 mm**
• Espesor:	**1.60 mm (FR4-TG155)**
• Cobre:	**1.0 oz (35 µm)**
• Acabado:	**ENIG (Electroless Gold)**
• Ranura Dieléctrica:	**Fresado 2.0 mm Edge**

TABLA DE TALADROS (DRILL):

• 4x Agujeros M2.5:	**Ø2.70 mm (Pad Ø5.0)**
• 2x Electrodos Choque:	**Ø4.20 mm M4 (Pad Ø8.0)**
• 2x Pines Buzzer:	**Ø1.00 mm (Pad Ø2.0)**
• 4x Pines Keystone 1048:	**Ø1.80 mm (Pad Ø3.6)**

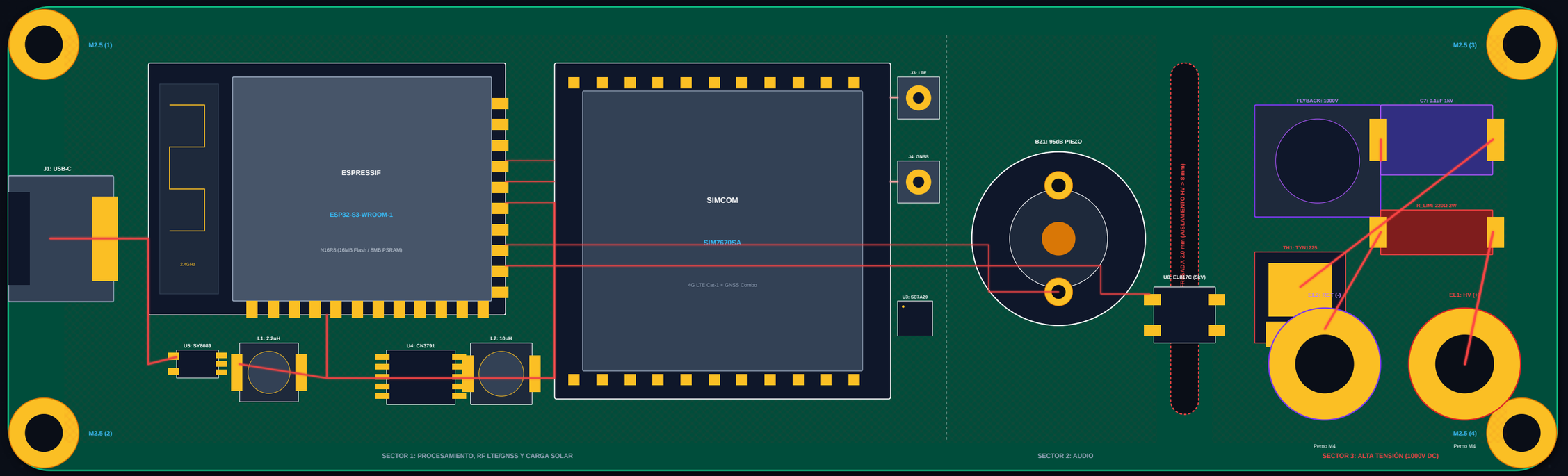
CONVENCIÓN DE CAPAS CAD:

- Top Layer Copper (Pistas Cara Superior F.Cu)
- Bottom Layer Copper (Pistas Cara Inferior B.Cu)
- Pads SMD / PTH (Oro Electroless ENIG)
- Serigrafía Blanca (Silkscreen F.SilkS / B.SilkS)

PROJECT: COLLARNET / COWAI V2 — VIRTUAL FENCE IOT			
DIAGRAMA DE PCB (LAYOUT COMPLETO ARTWORK TOP Y BOTTOM)		CLASE: **IPC-2221 / JLCPCB 2-L**	
DIMENSIONES: **110.60 x 33.10 mm**		REVISIÓN: **REV 2.2**	HOJA: **2 DE 6**
DISEÑADOR: **Ing. Hardware CowAI**	FECHA: **2026-10-27**	SUSTRATO: **FR4 1.6mm 1oz ENIG**	ESTATUS: **READY FOR SMT**

COLLARNET V2 — CARA SUPERIOR AISLADA (TOP LAYER COPPER + PADS SMT + SERIGRAFÍA BLANCA)

DIMENSIONES: 110.60 × 33.10 mm



COLLARNET V2 — CARA INFERIOR AISLADA (BOTTOM LAYER COPPER + CLIPS BATERÍA 21700 + NANO-SIM)

DIMENSIONES: 110.60 × 33.10 mm

